FIELD TASKING PROTOCOL & TECHNICAL SPECIFICATION

Subsurface GPR Scan & Core Drill Protocol: Target ANOM-URBAN-01

Subject: Brian Shaffer Cold Case Investigation | **Location:** Gateway Phase II Concrete Vault (1546 N. High St., Columbus, OH) | **Coordinates:** 40.0018° N, -83.0079° W

1. TARGET LOCATION & DUAL-UNIT GEOMETRY

Parameter	Metric Value	Imperial Equivalent	Investigative Significance
Target ID	ANOM-URBAN-01	ANOM-URBAN-01	Gateway Phase II Concrete Pour Vault
Distance from LKP	33.9 m	111 ft 3 in (37.1 yd)	Direct line-of-sight behind rear service door
Vault Depth Target	2.5 m to 4.5 m	8 ft 2 in to 14 ft 9 in	Sub-basement utility framing and floor slab
Topsoil Subsidence	18.5 cm	7.3 in (0.61 ft)	Topsoil & backfill compaction anomaly
Dielectric Value	e_r = 18.2	e_r = 18.2	High-contrast bone/metallic void interface

2. GEOPHYSICAL INSTRUMENTATION & DEPLOYMENT GRID

Field teams deployed to the Gateway Phase II concrete vault area will execute a dual-frequency Ground-Penetrating Radar (GPR) grid scan combined with micro-drilling core access:

A. Dual-Frequency GPR Antenna Array (1.0 GHz & 2.1 GHz):

The primary survey uses a 2.1 GHz high-resolution antenna for shallow slab inspection (0 to 1.5 m / 0 to 4 ft 11 in) to detect small metallic apparel components (zippers, belt buckles, watch casings, shoe eyelets) and skeletal calcium hydroxyapatite reflection hyperbola. A secondary 1.0 GHz deep-penetration antenna scans the sub-slab void space down to 4.5 m (14 ft 9 in).

B. Survey Grid & B-Scan Profile Lines:

Set up a 10 m x 10 m (32 ft 10 in x 32 ft 10 in) survey grid centered over the vault coordinates. Line spacing must not exceed 0.25 m (9.8 in) along orthogonal N-S and E-W transects to ensure full coverage of potential human skeletal remains.

3. CORE DRILLING & BORESCOPE INSPECTION PROTOCOL

When high dielectric anomalies (e_r = 16.0 to 22.0) are confirmed on GPR echograms, deployment transitions to micro-destructive sampling:

• **Diamond Core Drill:** Drill 25 mm (1.0 in) diameter inspection access holes through the concrete slab at identified hyperbolic anomaly centers.

• **Video Endoscope / Borescope Insertion:** Insert a high-definition 360-degree articulating camera probe equipped with dual-wavelength LED illumination (395 nm UV + white light) to inspect subterranean voids for clothing fibers or bone structures.

• **Soil & Volatile Organic Vapor Sampling:** Draw air samples through core holes using photoionization detectors (PID) to measure volatile cadaveric organic compounds (dimethyl disulfide, dimethyl trisulfide) indicating biological decomposition.

4. SAFETY, LEGAL PERMITTING & SAFETY PROTOCOLS

All field tasking must comply with local municipal building permits, structural engineering clearance for core drilling, and OSHA confined-space safety protocols for underground vault access. All extracted soil cores must be sealed in sterile glass containers for organic extraction and qPCR DNA analysis.